

MOUNT WILSON, CA, USA

WMO: 722890

Lat: 34.232N

Lon: 118.074W

Elev: 1739

StdP: 82.10

Time Zone: -8.00 (NAP)

Period: 82-15

WBAN: 93136

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.2	-2.8	-27.8	0.4	11.1	-25.0	0.5	10.9	11.1	2.7	9.4	3.6	2.9	320	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	31.2	15.3	29.9	15.1	28.1	14.3	17.8	27.1	16.9	26.7	16.2	26.0	2.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.6	12.9	21.7	13.0	11.6	21.6	12.2	10.9	21.9	57.4	27.0	54.6	26.9	52.0	25.8	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.6	5.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	8.3	8.3	8.7	11.2	15.0	18.4	22.6	22.2	20.9	15.2	10.3	6.8		
	DBStd	7.36	4.87	5.37	5.14	5.42	5.32	4.63	2.97	3.29	4.03	5.16	5.09	5.14		
	HDD10.0	546	89	89	88	51	21	4	0	0	2	17	59	127		
	HDD18.3	2046	309	282	300	220	127	52	4	7	23	121	242	358		
	CDD10.0	2012	38	41	47	86	175	256	389	379	327	178	69	28		
	CDD18.3	469	0	0	0	4	23	54	135	127	99	24	2	0		
	CDH23.3	3311	0	1	1	22	178	385	992	972	629	123	8	0		
(13)	CDH26.7	824	0	0	0	1	19	92	259	272	165	16	0	0		
(14)	Wind	WSAvg	2.3	2.4	2.7	2.6	2.6	2.1	1.9	1.9	2.2	2.1	2.5	2.5		
(15) Precipitation	PrecAvg	502	109	123	63	36	14	4	1	1	6	20	41	77		
	PrecMax	950	459	426	276	84	134	30	13	3	56	168	108	284		
	PrecMin	206	1	1	0	7	0	0	0	0	0	0	7	7		
	PrecStd	218	119	108	70	24	26	8	3	1	12	33	27	64		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	22.2	22.5	25.9	28.7	32.1	32.3	32.8	32.3	27.9	24.5	21.4		
		MCWB	7.7	8.7	9.9	11.1	12.8	14.9	16.2	16.8	14.8	12.2	9.8	8.7		
	2%	DB	18.4	19.3	20.1	22.7	27.2	28.8	30.9	31.2	30.1	26.2	21.7	17.9		
		MCWB	6.3	7.4	8.9	9.9	13.0	13.7	15.8	15.5	14.6	11.9	8.7	6.7		
	5%	DB	16.9	17.3	17.7	21.0	25.4	27.1	29.1	29.5	28.2	24.3	19.2	16.2		
		MCWB	5.6	6.9	8.1	9.3	12.1	13.3	15.3	15.3	14.1	11.0	8.1	5.6		
	10%	DB	14.7	15.4	15.9	19.2	23.3	25.3	27.7	27.8	27.1	22.4	17.2	13.8		
		MCWB	4.3	5.6	6.7	8.6	11.4	12.5	14.7	14.5	13.5	9.9	6.4	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.2	11.6	13.0	15.8	17.2	19.0	18.6	18.2	15.0	10.8	10.4		
		MCDB	16.6	17.2	18.8	21.4	25.3	28.0	27.7	26.9	26.2	23.8	20.1	16.9		
	2%	WB	7.6	8.8	10.2	11.5	14.4	15.7	17.9	17.7	16.8	13.4	10.0	8.8		
		MCDB	15.1	16.5	17.8	19.5	23.9	26.3	27.2	27.4	26.1	22.5	19.2	14.8		
	5%	WB	6.5	7.7	8.8	10.4	13.3	14.7	17.0	16.8	15.7	12.2	8.8	7.3		
		MCDB	13.4	15.7	16.7	19.0	23.3	24.8	26.6	26.8	25.3	21.2	17.5	12.0		
	10%	WB	5.5	6.6	7.5	9.3	12.4	13.5	16.2	16.0	14.8	11.0	7.7	5.9		
		MCDB	12.3	13.9	14.7	17.9	22.2	23.8	25.8	25.9	24.7	20.3	14.7	11.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.2	8.4	8.9	10.9	8.6	8.3	9.1	9.0	9.0	8.0	7.9		
		MCDBR	10.2	10.5	11.7	10.4	12.2	10.3	9.8	10.4	11.4	10.7	10.6	10.0		
	5% WB	MCWBR	6.4	7.3	8.5	7.5	8.8	7.8	7.7	7.1	7.2	7.0	6.7	7.0		
		MCWBR	8.9	9.7	10.7	10.6	11.8	10.4	8.7	9.1	10.0	10.1	9.9	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.267	0.282	0.287	0.288	0.301	0.303	0.292	0.282	0.284	0.253	0.252		
		taud	2.618	2.619	2.551	2.539	2.569	2.531	2.579	2.602	2.626	2.588	2.683	2.672		
	Ebn at Noon		974	998	1000	1000	994	976	972	983	986	962	977	967		
		Edh at Noon	74	82	96	102	101	104	99	95	98	84	69	66		
(44) All-Sky Solar Radiation	RadAvg		3.17	4.06	5.48	6.71	7.39	7.92	7.68	7.26	6.16	4.67	3.56	2.88		
	RadStd		0.23	0.42	0.30	0.32	0.41	0.41	0.36	0.21	0.19	0.29	0.18	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.64	N/A	-1.66	+0.71	N/A	N/A	N/A	-114	+225	+110

Nomenclature: See separate page